# Lecture 1.2: Variables, Data Types, and `typeof`

## Variables in JavaScript

A **variable** is a named container for a value. In your lecture you used `var` to declare variables.

### Declaring variables

- `var` (older, function-scoped)
- `let` (modern, block-scoped)
- `const` (modern, block-scoped, cannot be reassigned)

> **Instructor tip:** For new code, prefer `let` and `const`. Learn `var` because you will see it in older code and tutorials.

## JavaScript is dynamically typed

In JavaScript, a variable can hold any type of value, and it can change type later.

### Your lecture example (idea)

```
// A single variable holding different kinds of values
var a = 5;
console.log(typeof a);

a = "Hello";
console.log(typeof a);

a = 34.5;
console.log(typeof a);

a = true;
console.log(typeof a);

a = [3, 6, 2, 6];
console.log(typeof a);
```



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```
a = undefined;
console.log(typeof a);
```

**Output:**

Typical outputs: `number`, `string`, `number`, `boolean`, `object`, `undefined`

## Important notes about types

### Arrays are reported as `object`

One common beginner surprise is:

```
typeof [1, 2, 3] // "object"
```

To specifically check arrays, use:

```
Array.isArray([1, 2, 3]) // true
```

### `null` and `typeof`

Another famous JavaScript quirk:

```
typeof null // "object"  (historical behavior)
```

### Common primitive types

| Type | Example | `typeof` result |
|---|---|---|
| Number | `10`, `3.14` | `"number"` |



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| Type | Example | `typeof` result |
|---|---|---|
| String | `"Hello"` | `"string"` |
| Boolean | `true`, `false` | `"boolean"` |
| Undefined | `undefined` | `"undefined"` |
| BigInt | `9007199254740993n` | `"bigint"` |
| Symbol | `Symbol("id")` | `"symbol"` |

## Practice tasks

1. Declare a variable and assign a number, then a string, then a boolean. Log `typeof` each time.
2. Create an array and check both `typeof` and `Array.isArray`.
3. Try `typeof null` and explain why it's confusing.

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*JS Refined Notes — Lecture 1.2*



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